

Computer Science & information Technology

Compiler Design

DPP: 1

Syntax Directed Translation & Intermediate Code Generation

& Code Optimisation

Q1 Synthesized attributes can be easily simulated using

- (A) LL grammar
- (B) LR grammar
- (C) Ambiguous grammar
- (D) None of these

Q2 Consider the following translation rules for the grammar G:

$S \rightarrow a \quad \{\text{print "A"}\} A$
 $A \rightarrow b \quad \{\text{print "C"}\} B$
 $A \rightarrow \epsilon \quad \{\text{print "C"}\}$
 $B \rightarrow e \quad \{\text{print "B"}\} A$
 $B \rightarrow \epsilon \quad \{\text{print "C"}\}$
 $C \rightarrow c \quad \{\text{print "A"}\}$

What will be the output for the input string "abebebe" you top-down parser?

- (A) ACBCCBAC
- (B) ACCBCCBC
- (C) ACBCCBCC
- (D) ACBCBCBC

Q3 Consider the following attribute grammar:

$A \rightarrow BA' \quad A' \cdot b = A \cdot a$
 $A \cdot a = A' \cdot b$
 $A_1' \rightarrow +BA_2' \quad A_2' \cdot b = A_1' \cdot b + B \cdot a$
 $A_1' \cdot a = A_2' \cdot a$

Which of the following is true?

- (A) Both a and b are inherited attributed.
- (B) Both a and b are synthesized attributed.
- (C) a is inherited, b is synthesized
- (D) b is inherited, a is synthesized

Q4 Consider the following grammar:

$E \rightarrow E + T \mid T$
 $T \rightarrow T / F \mid F$

$F \rightarrow F * A \mid A$

$A \rightarrow \text{id}$

Which one of the following is/are true?

- (A) / have higher precedence than *
- (B) * have higher precedence than +
- (C) + have lower precedence than /
- (D) *, +, / all have same precedence.

Q5 A shift reduce parser perform action specified within process immediately after reduction to the corresponding rule of grammar.

$S \rightarrow abV \quad \{\text{print '11'}\}$
 $S \rightarrow cc \quad \{\text{print '22'}\}$
 $V \rightarrow Sd \quad \{\text{print '33'}\}$

What is the translation of "ababccdd" using the SDT scheme described by above rules?

- (A) 2233113311
- (B) 1133113322
- (C) 2211331122
- (D) 1122113322

Q6 Consider the following transition rules:

$A \rightarrow BC$
 $C \rightarrow +BC \mid A + \mid \epsilon$
 $B \rightarrow DB \quad \{\text{print '+'}\} \mid \epsilon$
 $D \rightarrow \text{id} \quad \{\text{print number value}\}$

If input is given "2 + 34" then his translation scheme will generate output.

- (A) 2 + 3 + 4 +
- (B) + 2 + 3 + 4
- (C) ++ 2 + 34
- (D) 2 + 34 ++

Q7 ____is performed by attaching rules or algorithms to production in a grammar.

- (A) Lexical analysis



- (B) Execution
 (C) Syntax directed translation
 (D) None of these.

Q8 Consider a translation scheme is given as:

$$S \rightarrow S_1 + S_2 \quad \{S \cdot \text{val} = S_1 \cdot \text{val} + S_2 \cdot \text{val}\}$$

$$S \rightarrow S_1 * S_2 \quad \{S \cdot \text{val} = S_1 \cdot \text{val} * S_2 \cdot \text{val}\}$$

$$S \rightarrow \text{id} \quad \{S \cdot \text{val} = \text{id}\}$$

What will be the output for $5 * 6 + 7$?

- (A) 18
 (B) 37
 (C) 65
 (D)

Cannot be identified because it is ambiguous grammar.

Q9 Consider the following SDT :

$$S \rightarrow E \quad \{S \cdot \text{val} = E \cdot \text{val}\}$$

$$E \rightarrow E + T \quad \{E \cdot \text{val}\} = E_1 \cdot \text{val} + T \cdot \text{val}\}$$

$$E \rightarrow T \quad \{E \cdot \text{val} = T \cdot \text{val}\}$$

$$T \rightarrow T \times F \quad \{T \cdot \text{val} = T_1 \cdot \text{val} * f \cdot \text{val}\}$$

$$T \rightarrow F \quad \{T \cdot \text{val} = f \cdot \text{val}\}$$

$$F \rightarrow (E) \quad \{F \cdot \text{val} = E \cdot \text{val}\}$$

$$F \rightarrow a \quad \{f \cdot \text{val} = a\}$$

What will be the output of the expression " $20 + 8 \times 6$ "



Answer Key

Q1 (B)

Q2 (D)

Q3 (D)

Q4 (B, C)

Q5 (A)

Q6 (D)

Q7 (C)

Q8 (D)

Q9 68



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Hints & Solutions

Q1 Text Solution:

Synthesized attributes can be easily simulated by LR grammar.

As Synthesized attribute are evaluated in bottom-up manner and LR grammar is also evaluated in bottom-up manner.

Q2 Text Solution:

After evaluating output will be ACBCBCBC. Therefore, option (d) correct.

Q3 Text Solution:

For the first production a, b are attributes, such that

$a \in A$, and $b \in A'$.

Now if you look into the translation closely,

$A' \cdot b = A \cdot a$ (b attribute is taking value from its parent attribute)

$A' \cdot a = A' \cdot b$ (a attribute is taking value from one of its child attributes)

Similarly for second production, b is taking values from its parent as well as sibling which left to it (it is L-attributed as well as S)

So, (b) follows inherited attribute definition while (a) follows synthesized attribute definition.

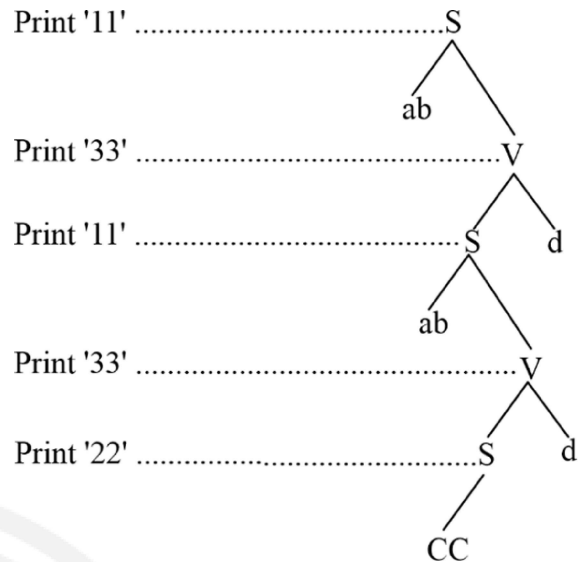
Q4 Text Solution:

*have higher precedence than +

+ have lower precedence than/

Q5 Text Solution:

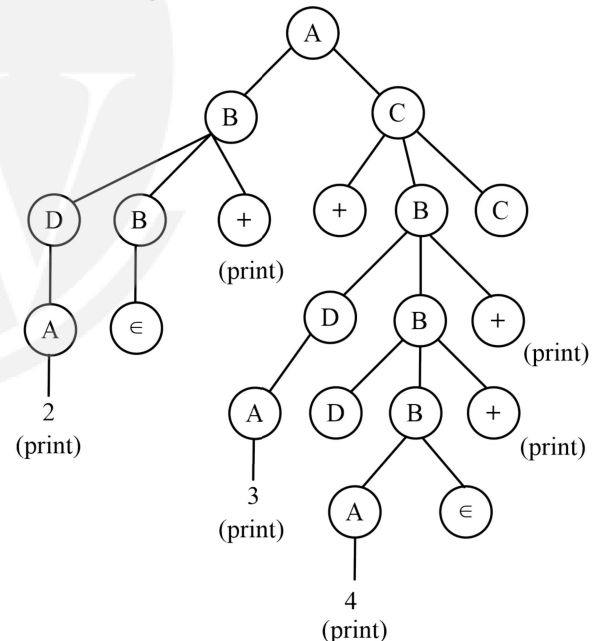
Given input string ababccdd



So, output will be 2233113311. Therefore option (a) is correct.

Q6 Text Solution:

The input given is "2 + 34"



On scanning them left to right.

Output = 2 + 34 ++

So, option (d) is correct.

Q7 Text Solution:


SDT is done by attaching augmented rules to the analysis. SDT rules used the following:

1. Lexical value of node
2. Constants
3. Attributes associated to the non-terminal in their definition.

Q8 Text Solution:

The given grammar is ambiguous grammar

$S \rightarrow S1 + S2 \mid S1 * S2 \mid id$ is an ambiguous grammar.
Because for expression $5 * 6 + 7$ two outputs are possible.

Output 1 : 37

Output 2 : 65

So option (d) is correct answer.

Q9 Text Solution:

Output will be $20 + 6 * 8 = 68$

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